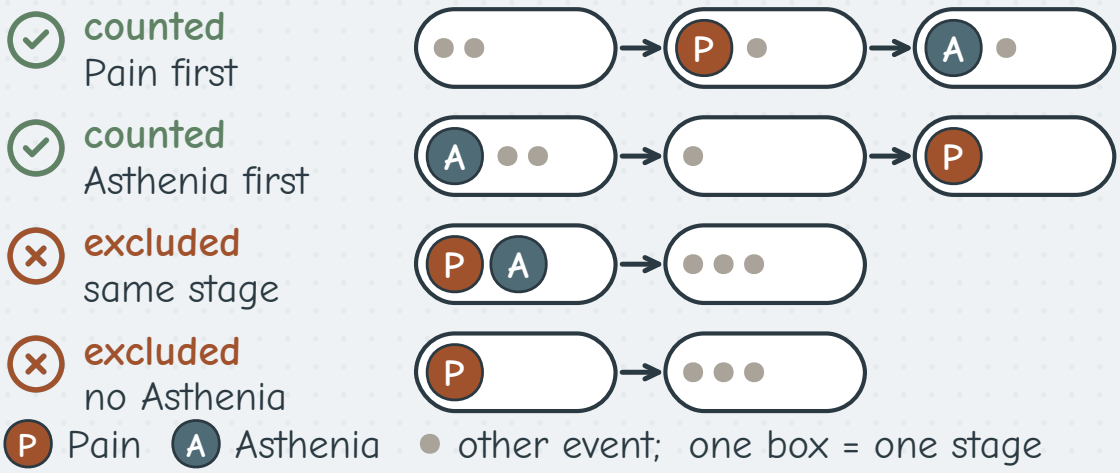
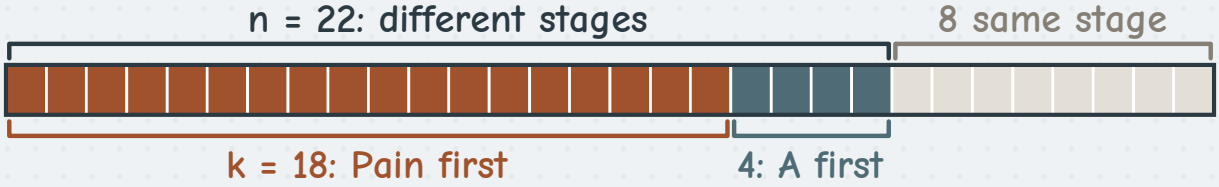
Claim: After Moderna, Pain typically occurs before Asthenia.

## 1 WHICH REPORTS COUNT?




983 Moderna reports; **30 mention both events:**




## 2 A ONE-QUERY BASELINE

"I count a report only if **Pain comes first**."

**n = 18, k = 18** → p < 0.0001 → supported

N = K!

The **4 Asthenia-first reports** vanish, so n = k.

## 3 OUR STAGED PIPELINE

**S1 Articulate**
both at different stages; null 0.5; exact binomial

→

**S2 Retrieve**
SQL under that rule returns
**n = 22, k = 18**

→

**S3 Test**
exact binomial
p = 0.002
**verdict: supported**